

8.1

$$A = \begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & -2 & 0 \\ -1 & 3-\lambda & -1 \\ 2 & 3 & 1-\lambda \end{vmatrix} \quad 2.5$$
$$= (-1) \cdot (-1)^5 \begin{vmatrix} -1-\lambda & -2 \\ 2 & 3 \end{vmatrix} + (1-\lambda) \cdot \begin{vmatrix} -1-\lambda & -2 \\ -1 & 3-\lambda \end{vmatrix}$$

$$= ((-1-\lambda) \cdot 3 + 4) + (1-\lambda) \cdot ((-1-\lambda) \cdot (3-\lambda) - 2)$$

$$= 1 - 3\lambda + (1-\lambda) \cdot (-3 + \lambda - 3\lambda + \lambda^2 - 2)$$

$$= 1 - 3\lambda + (1-\lambda) \cdot (\lambda^2 - 2\lambda - 5)$$

$$= 1 - 3\lambda + \lambda^2 - 2\lambda - 5 - \lambda^3 + 2\lambda^2 + 5\lambda$$

$$= -\lambda^3 + 3\lambda^2 - 4$$

$$= \lambda^3 - 3\lambda^2 + 4$$

$$= (\lambda - 2)^2 \cdot (\lambda + 1)$$

$$\lambda_1 = 2$$

$$\lambda_2 = -1$$

↳ algebraischer
Mehrfachheit 2

↳ algebraischer
Mehrfachheit 1

$$M_2 = \ker \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix} = \ker \begin{pmatrix} 0 & -5 & 3 \\ -1 & 1 & -1 \\ 0 & 5 & -3 \end{pmatrix} = \ker \begin{pmatrix} -1 & 1 & -1 \\ 0 & -5 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$M_2 = \text{Span} \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \right\}$$

$$M_1 = \text{Ker} \begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 2 & 3 & 2 \end{pmatrix} = \text{Ker} \begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 0 & 11 & 0 \end{pmatrix} = \text{Ker} \begin{pmatrix} -1 & 4 & -1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$M_1 = \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\} \quad w = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\lambda_1 = 2 \text{ má' algebraickou násobnosť } 2, \text{ ale } M_2 = \text{Span} \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \right\}$$

↳ Jordanov retízel

$$(A - 2I) V_2 = V_1$$

$$\begin{matrix} \nearrow \\ (-3) \\ \searrow \end{matrix} \begin{pmatrix} -3 & -2 & 0 & | & -2 \\ -1 & 1 & -1 & | & 3 \\ 2 & 3 & -1 & | & 5 \end{pmatrix} \sim \begin{pmatrix} 0 & -5 & 3 & | & -11 \\ -1 & 1 & -1 & | & 3 \\ 0 & 5 & -3 & | & 11 \end{pmatrix} \sim \begin{pmatrix} -1 & 1 & -1 & | & 3 \\ 0 & -5 & 11 & | & -11 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} -2 \\ 0 \\ -1 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}$$

vždy rovný U , pro které platí

$$f_A(U) = U \quad \rightarrow \text{invariantní rovinou}$$

$$\hookrightarrow \forall x \in U: Ax \in U$$

- platí, že lineární obal spojený libovolného počtu jordanových řetězců nějakého operátoru je jeho podprostor

Jd' má'm dva řetězky: $v_1 \rightarrow v_2$ (délka 2)
 $\nwarrow w$ (délka 1)

$$\mathbb{R}^3 = \text{Span} \left\{ \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\} \oplus \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\mathbb{R}^3 = \text{Span} \{ v_1, v_2 \} \oplus \text{Span} \{ w \}$$

platí: $(A - 2I)v_1 = 0$

$$\bullet Av_1 = 2v_1$$

$$(A - 2I)v_2 = v_1$$

$$Av_2 - 2v_2 = v_1$$

$$\bullet Av_2 = v_1 + 2v_2$$

$$(A + I)w = 0$$

$$\bullet Aw = -w$$

$$- U_1 = \text{Span} \{v_1, v_2\} \quad \checkmark$$

$$Av_1 = 2v_1 \quad \in U$$

$$Av_2 = v_1 + 2v_2 \quad \in U$$

$\checkmark \rightarrow$ invariantni

$$- U_2 = \text{Span} \{v_1, w\} \quad \checkmark$$

$$Av_1 = 2v_1 \quad \in U$$

$$Aw = -w \quad \in U$$

$\checkmark$

$$U_3 = \text{Span} \{v_2, w\}$$

$$Av_2 = v_1 + 2v_2 \notin U$$

$$Aw = -w$$

$\times$

Teo 2 invariantni roing

$$U_1 = \text{Span} \{v_1, v_2\}$$

$$B_{U_1} = (v_1, v_2)$$

$$[t_A|_{U_1}]_{B_{U_1}}^{B_{U_1}} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$U_2 = \text{Span} \{v_1, w\}$$

$$B_{U_2} = (v_1, w)$$

$$[t_A|_{U_2}]_{B_{U_2}}^{B_{U_2}} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$